

PVsyst - Simulation report

Grid-Connected System

Project: GRID CONNECTED WITH SHADING

Variant: New simulation variant

Sheds on ground

System power: 410 kWp

George Town - India



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PVsyst V7.4.7

VC0, Simulation date:
25/03/26 20:45
with V7.4.7

Project summary

Geographical Site

George Town

India

Situation

Latitude 13.09 °N

Longitude 80.28 °E

Altitude 24 m

Time zone UTC+5.5

Project settings

Albedo 0.20

Weather data

George Town

Meteonorm 8.1 (1996-2015) - Synthetic

System summary

Grid-Connected System

PV Field Orientation

Fixed plane

Tilt/Azimuth 15 / 20 °

Sheds on ground

Near Shadings

Linear shadings : Fast (table)

User's needs

Unlimited load (grid)

System information

PV Array

Nb. of modules

630 units

Pnom total

410 kWp

Inverters

Nb. of units

1 unit

Pnom total

360 kWac

Pnom ratio

1.138

Results summary

Produced Energy 622518 kWh/year Specific production 1520 kWh/kWp/year Perf. Ratio PR 80.61 %

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General parameters

Grid-Connected System

PV Field Orientation

Orientation

Fixed plane
Tilt/Azimuth 15 / 20 °

Sheds on ground

Sheds configuration

Nb. of sheds 66 units

Sizes

Sheds spacing 5.65 m
Collector width 2.63 m
Ground Cov. Ratio (GCR) 46.5 %
Top inactive band 0.02 m
Bottom inactive band 0.02 m

Shading limit angle

Limit profile angle 12.5 °

Models used

Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

Horizon

Free Horizon

Near Shadings

Linear shadings : Fast (table)

User's needs

Unlimited load (grid)

PV Array Characteristics

PV module

Manufacturer AE Solar
Model AE 650ME-132
(Original PVsyst database)

Unit Nom. Power 650 Wp
Number of PV modules 630 units
Nominal (STC) 410 kWp
Modules 35 string x 18 In series

At operating cond. (50°C)

Pmpp 375 kWp
U mpp 617 V
I mpp 607 A

Total PV power

Nominal (STC) 410 kWp
Total 630 modules
Module area 1957 m²
Cell area 1834 m²

Inverter

Manufacturer SolarMax
Model Solarmax 360TS-SV Multi MPPT
(Original PVsyst database)

Unit Nom. Power 360 kWac
Number of inverters 1 unit
Total power 360 kWac
Operating voltage 510-800 V
Pnom ratio (DC:AC) 1.14
Power sharing within this inverter

Total inverter power

Total power 360 kWac
Number of inverters 1 unit
Pnom ratio 1.14

Array losses

Thermal Loss factor

Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses

Global array res. 17 mΩ
Loss Fraction 1.5 % at STC

Module Quality Loss

Loss Fraction -0.8 %

Module mismatch losses

Loss Fraction 2.0 % at MPP

Strings Mismatch loss

Loss Fraction 0.2 %

IAM loss factor

Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.962	0.892	0.816	0.681	0.440	0.000



Project: GRID CONNECTED WITH SHADING

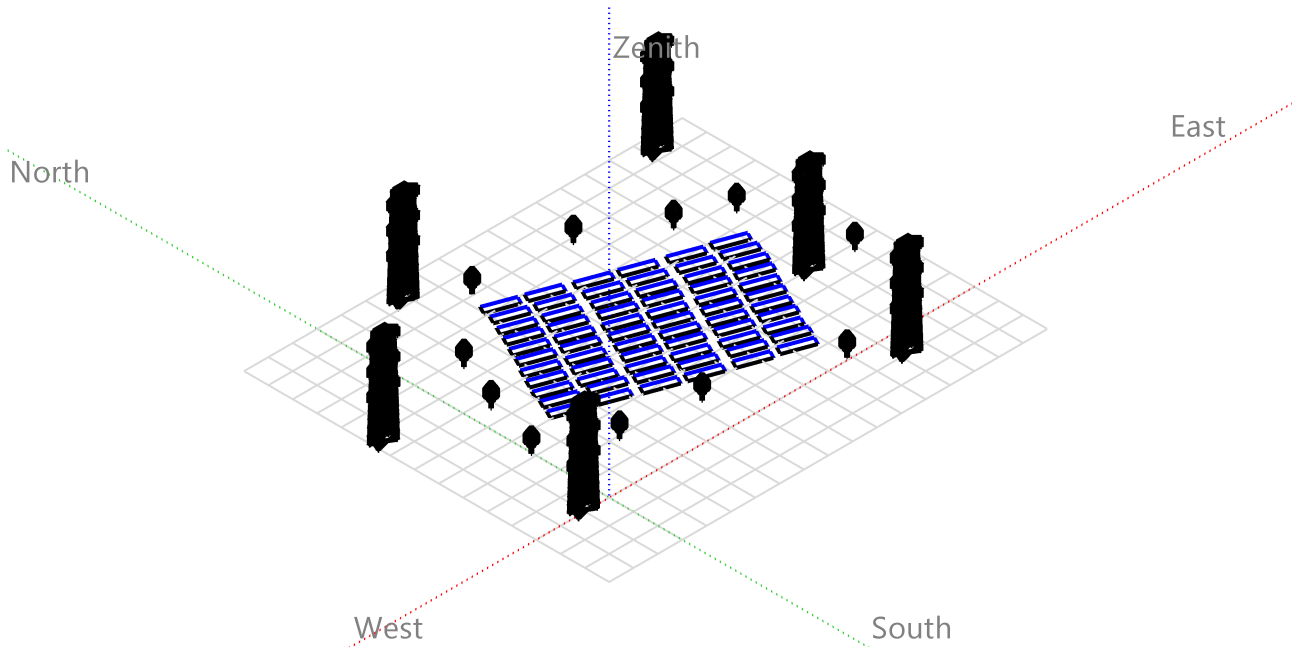
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Near shadings parameter

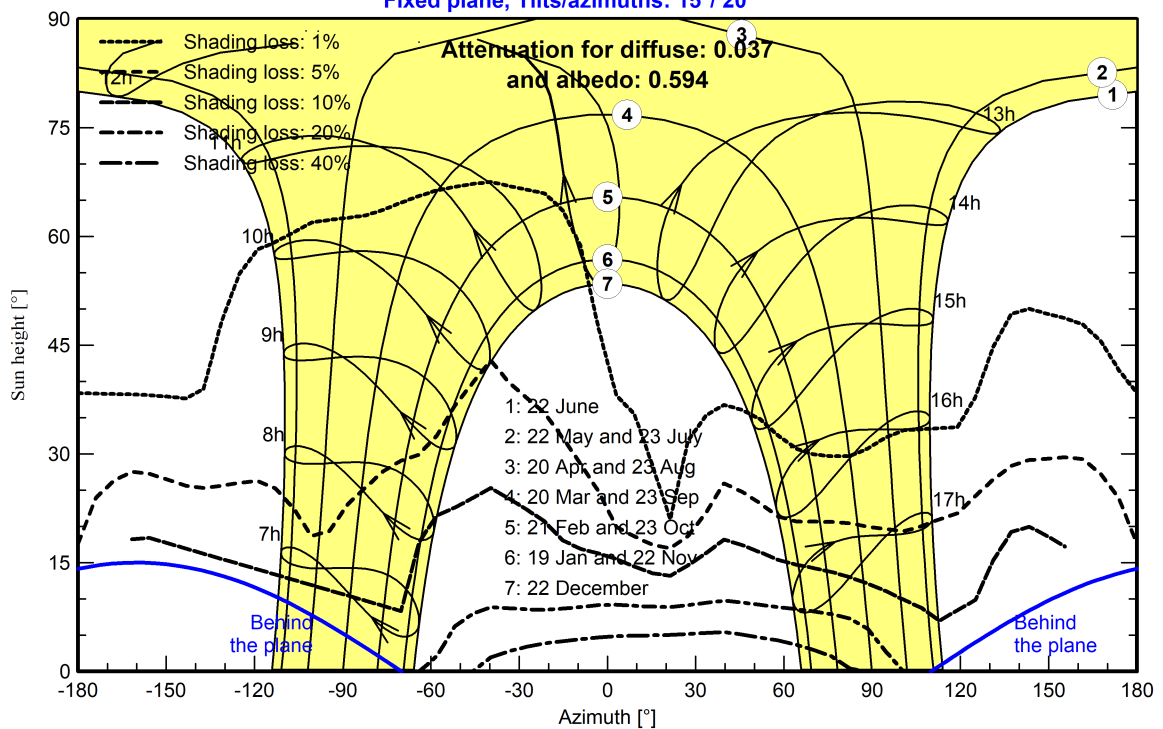
Perspective of the PV-field and surrounding shading scene



Iso-shadings diagram

Orientation #1

Fixed plane, Tilts/azimuths: 15°/ 20°





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Main results

System Production

Produced Energy

622518 kWh/year

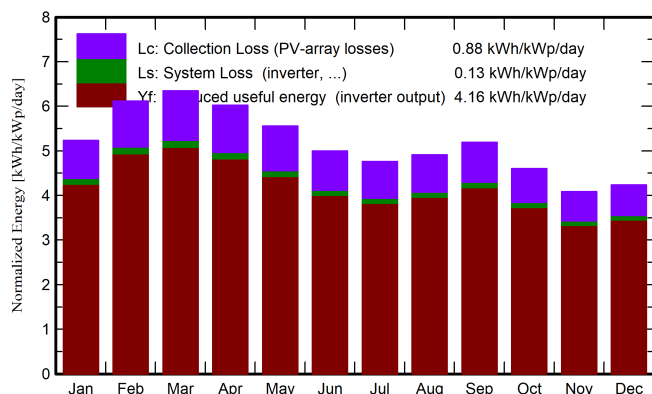
Specific production

1520 kWh/kWp/year

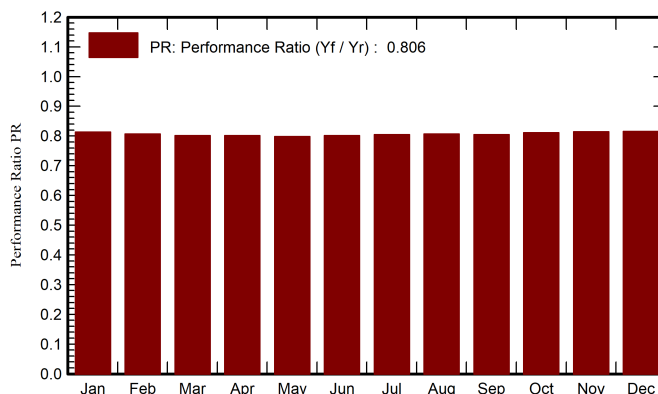
Perf. Ratio PR

80.61 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor	DiffHor	T_Amb	GlobInc	GlobEff	EArray	E_Grid	PR
	kWh/m²	kWh/m²	°C	kWh/m²	kWh/m²	kWh	kWh	ratio
January	144.2	66.32	25.11	162.4	154.4	55703	54090	0.813
February	156.4	60.47	26.21	171.4	164.2	58356	56619	0.807
March	189.7	74.76	28.44	196.7	189.4	66534	64532	0.801
April	184.7	81.64	30.16	180.7	173.9	61078	59278	0.801
May	182.6	91.33	32.17	172.3	165.3	57941	56285	0.798
June	162.3	84.49	31.21	150.0	143.5	50692	49248	0.802
July	158.0	91.51	30.64	147.6	140.8	50055	48645	0.805
August	157.1	89.73	29.68	152.3	146.0	51827	50330	0.807
September	153.7	78.09	28.85	155.7	149.2	52861	51316	0.805
October	134.9	74.42	27.86	142.8	136.9	48862	47421	0.811
November	112.4	62.00	25.92	122.7	116.8	42149	40914	0.815
December	117.8	63.02	25.21	131.4	124.6	45150	43841	0.815
Year	1853.9	917.78	28.47	1885.9	1804.8	641208	622518	0.806

Legends

GlobHor Global horizontal irradiation

DiffHor Horizontal diffuse irradiation

T_Amb Ambient Temperature

GlobInc Global incident in coll. plane

GlobEff Effective Global, corr. for IAM and shadings

EArray Effective energy at the output of the array

E_Grid Energy injected into grid

PR Performance Ratio



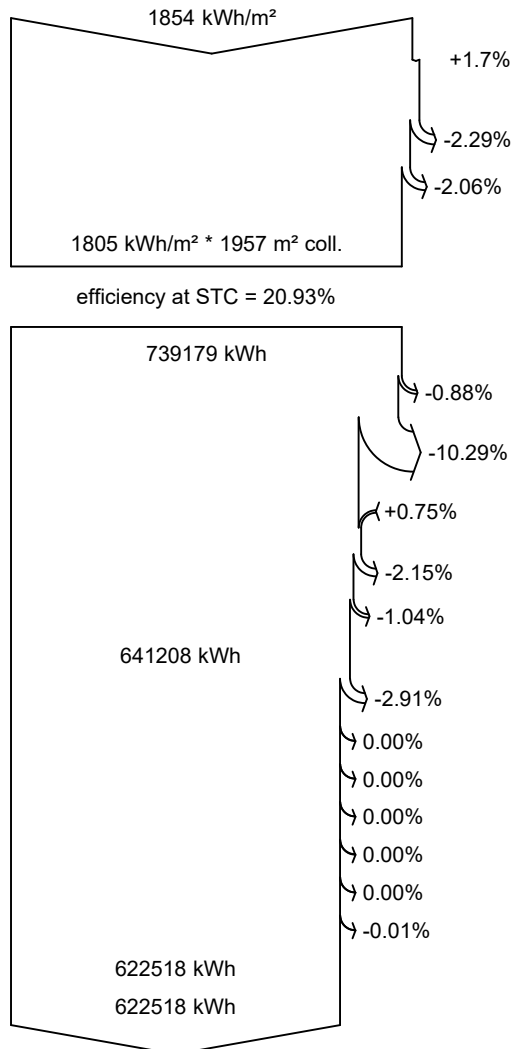
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Loss diagram



Global horizontal irradiation

Global incident in coll. plane

Near Shadings: irradiance loss

IAM factor on global

Effective irradiation on collectors

PV conversion

Array nominal energy (at STC effic.)

PV loss due to irradiance level

PV loss due to temperature

Module quality loss

Mismatch loss, modules and strings

Ohmic wiring loss

Array virtual energy at MPP

Inverter Loss during operation (efficiency)

Inverter Loss over nominal inv. power

Inverter Loss due to max. input current

Inverter Loss over nominal inv. voltage

Inverter Loss due to power threshold

Inverter Loss due to voltage threshold

Night consumption

Available Energy at Inverter Output

Energy injected into grid

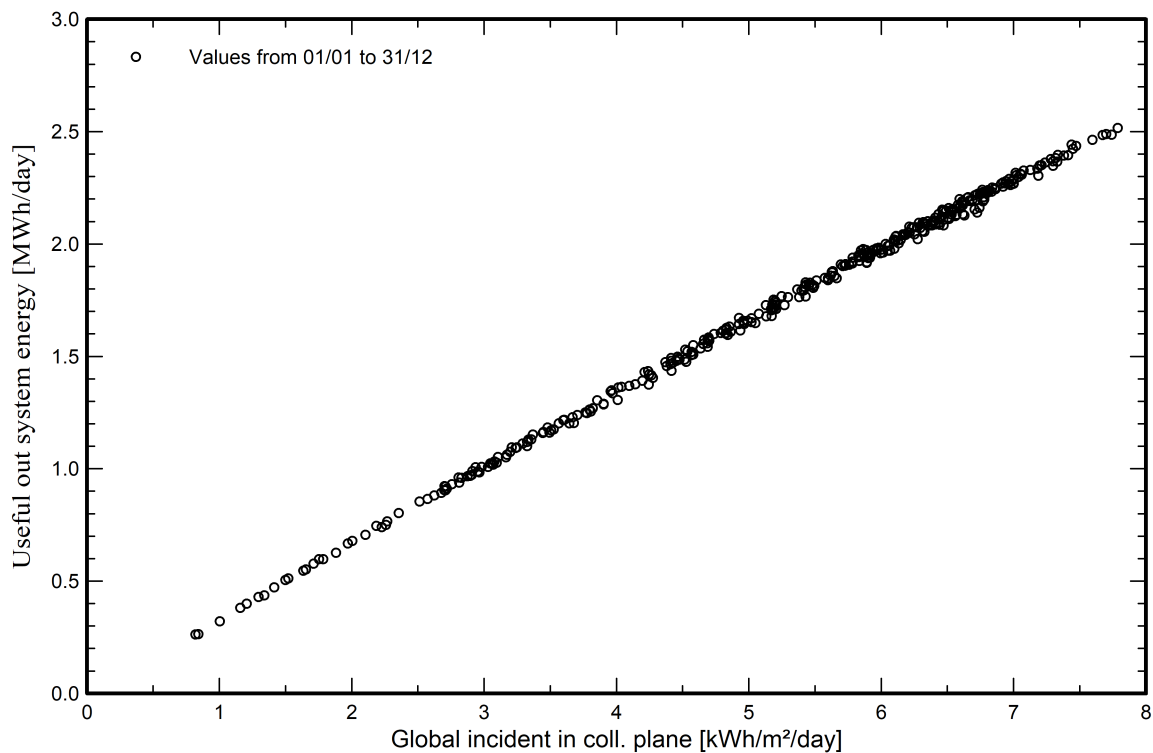


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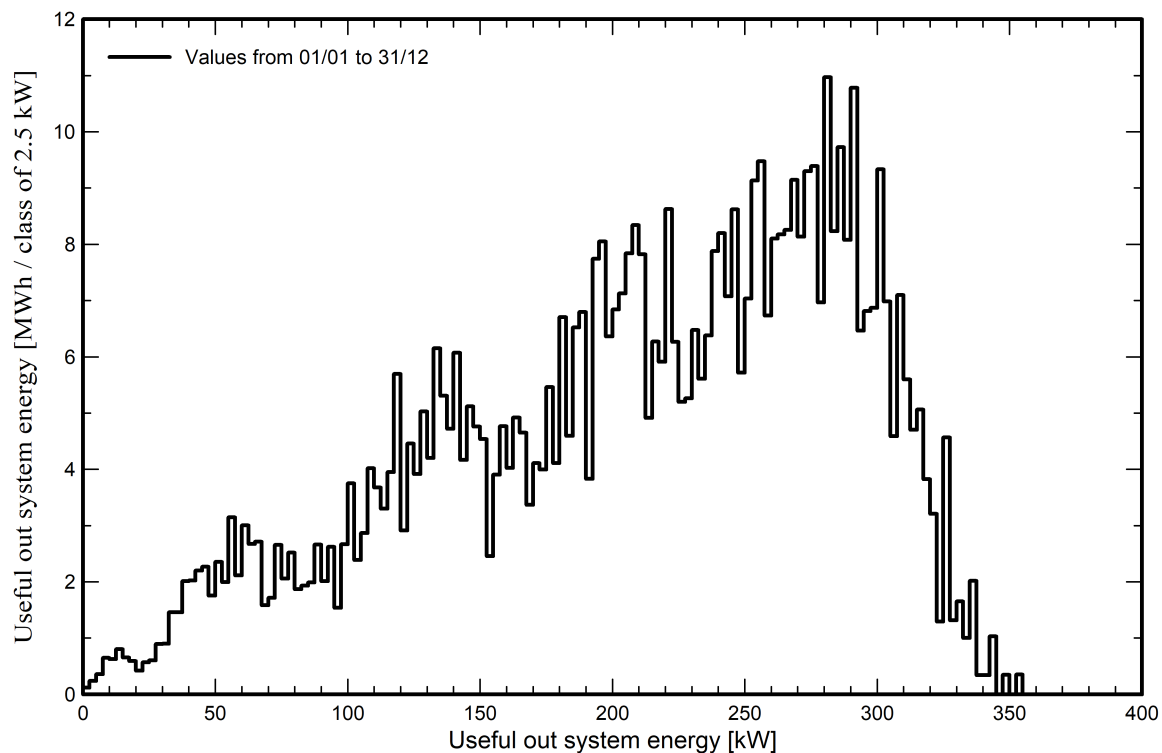
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Predef. graphs

Daily Input/Output diagram



System Output Power Distribution

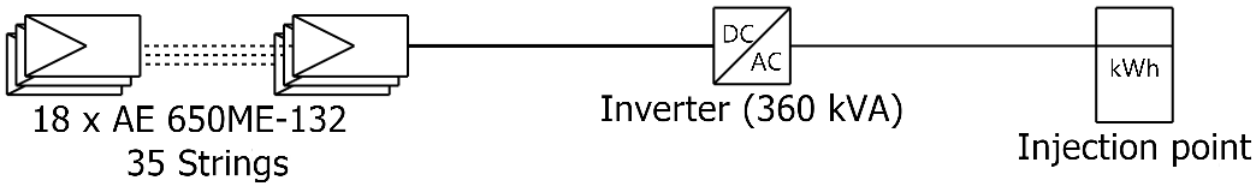




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Single-line diagram



PV module	AE 650ME-132
Inverter	Solarmax 360TS-SV Multi MPPT
String	18 x AE 650ME-132

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